

Introduction

Acute Myeloid Leukemia (LAML; TCGA acronym) is a cancer in which the bone marrow produces an abundant amount of blood cells. This cancer is the most common Acute Leukemia in Adults and will get worse rapidly if treatments aren't administered [1]. LAML is often fatal as the 5-year survival rate is about 27% for adults [10]. Drug treatments are also limited, and the development process of new drugs is lengthy. It is therefore imperative to consider the repurposing of existing drugs [11].

Precision oncology is a field that aims to provide personalized treatments for cancer patients based on their specific genetic and molecular profiles [4]. Personalized drug screening is a critical component of precision oncology as it helps identify drugs that are most effective for a particular patient's cancer [5]. This study aimed to investigate the potential of repurposing drugs through personalized drug screening with a focus on LAML.

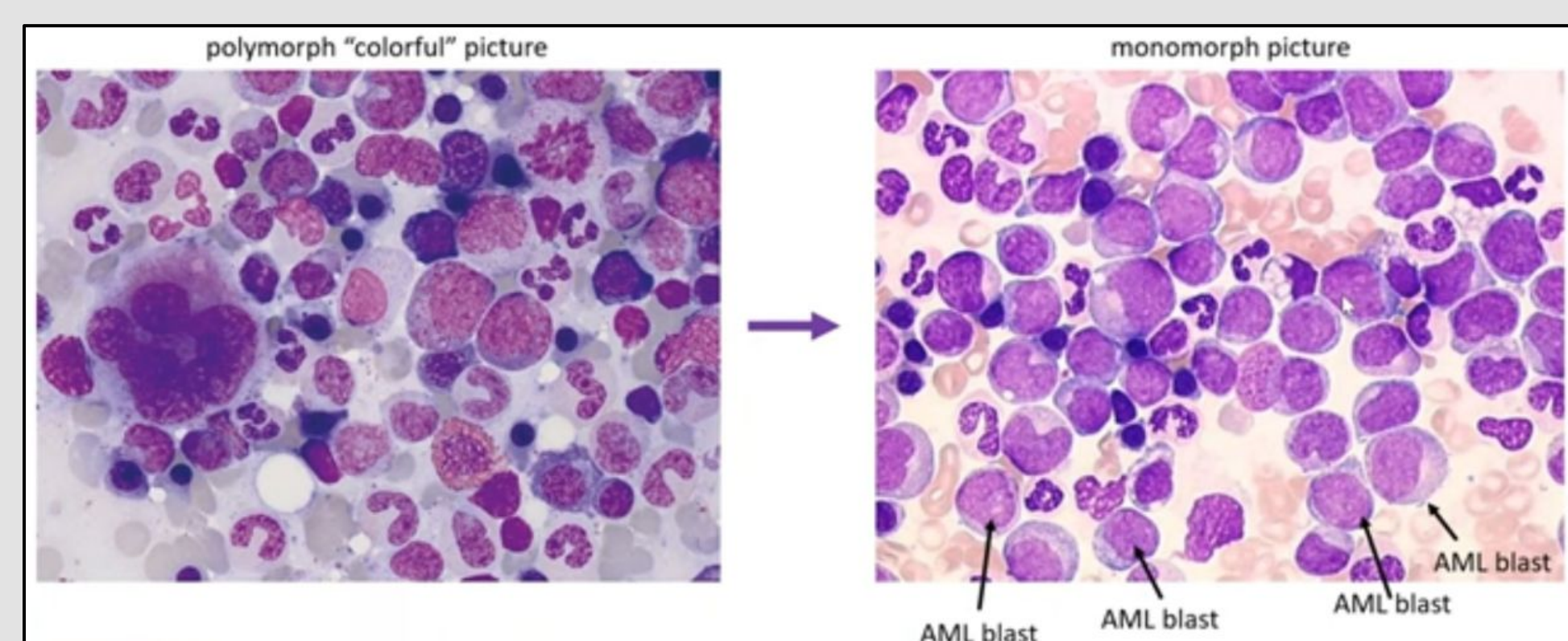


Figure 1: Representative images of blood cell population from normal and Acute Myeloid Leukemia [3]. Taken from *Leukemia & Lymphoma Society of Canada*.

The length of time required for traditional drug screening can range from a couple of days to weeks, and availability of drug perturbation screening facilities is limited in most clinics, leading to logistical challenges and further delays. Model-based virtual screening has emerged as a potential solution to address these challenges. Drug screening is increasingly becoming important, and historical data from personalized drug screening of other patients can be used to build models for treatment recommendations. The objective of this research is to utilize a computational model trained on cancer cell lines, combining multi-cancer drug data features, to predict the perturbations of an unknown drug of the specific cancer type Acute Myeloid Leukemia (LAML).

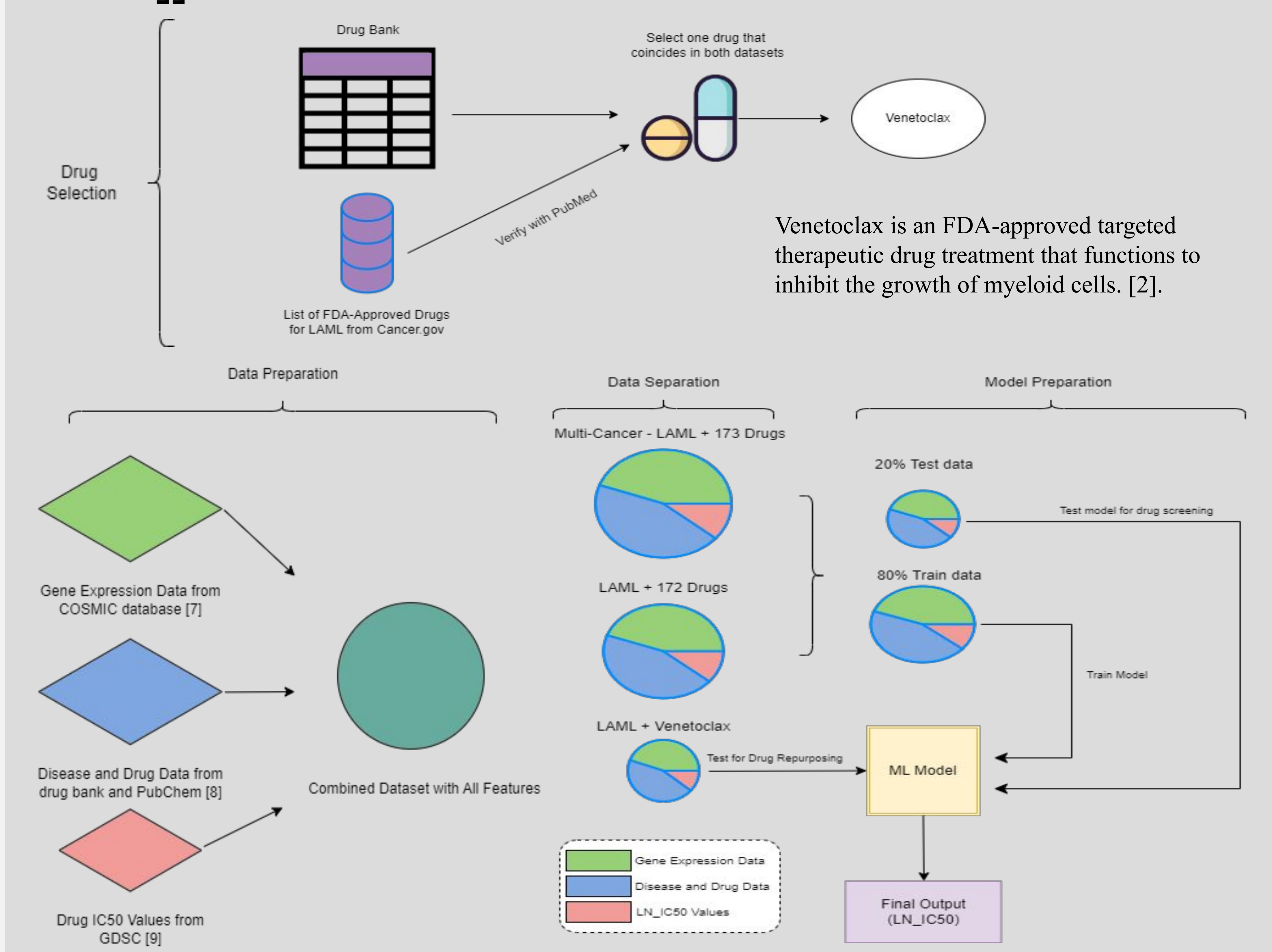


Figure 2: Drug Screening Process Illustration [6]

Data Description

- 173 Drugs from Drug Bank and PubChem
- 3 FDA Approved Drugs for LAML
- 135,242 Perturbation information on cancer cell lines

Methodology



Results

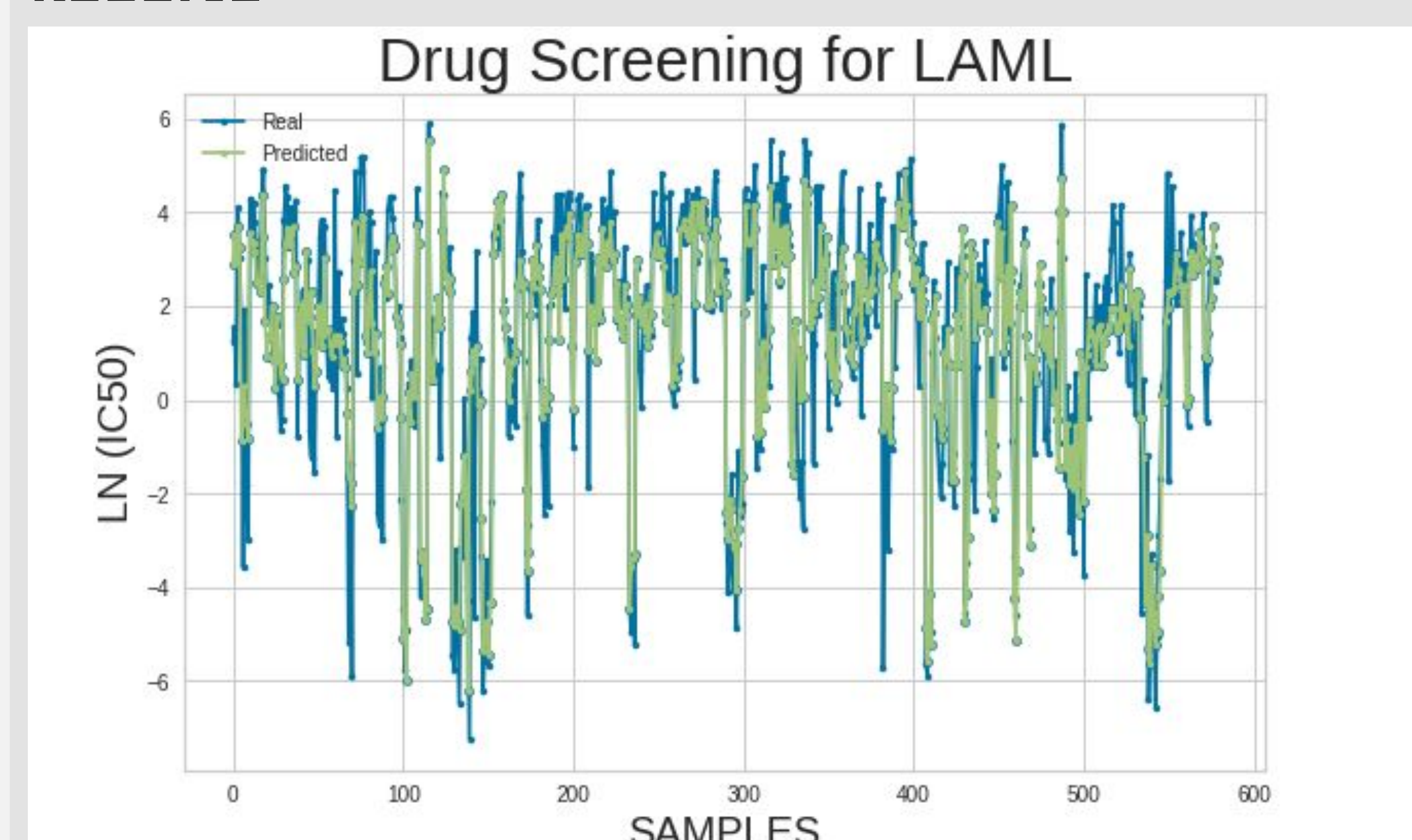


Figure 3: A graphical representation of the model being tested for LAML. The graph shows a strong correlation between real and predicted IC50 values.

Table 1 : The table shows the evaluation result of the training and test data. The best regressor for this experiment is the Light Gradient Boosting Machine (LightGBM).

MAE	MSE	
0.8839	1.3864	0.8271

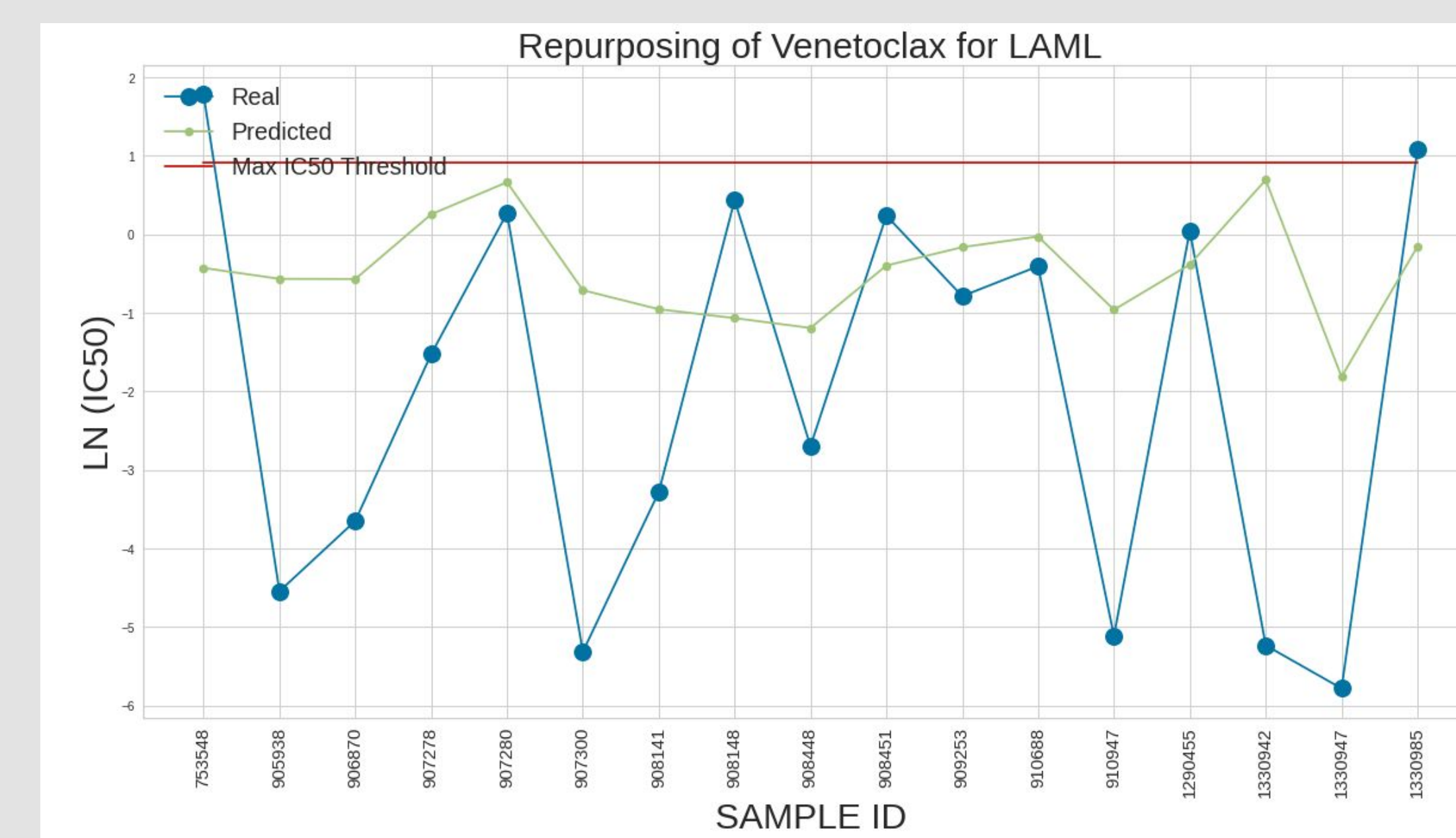


Figure 4: Graphical representation of predicted and real Venetoclax efficacy in individual patients.

Table 2 : The evaluation metrics of repurposing model, however it is not the only method of validation as it's important to also consider the Maximum IC50 Threshold

MAE	MSE	
2.2785	7.9658	0.2891

Discussion/Conclusion

The results showed that the best model, LightGBM, was capable of accurately predicting drug perturbations on unseen samples. The Mean Absolute Error (MAE) and R-Squared were 0.8839 and 0.8271, respectively. A MAE value less than 1.0, indicates a very small error when predicting real IC50 values. An R-Squared value > 0.7 , indicates a strong correlation whereas an R-squared < 0.3 , indicates a weak correlation. The drug screening graph showed a strong correlation for LAML.

For the drug repurposing of Venetoclax, the R-Squared was about 0.289, indicating a weak correlation. However, when analyzing another validation, where the maximum IC50 threshold was viewed, more accurate predictions for 88% of the samples were observed.

This research highlights the potential of Natural Language Processing-based drug features in personalized drug screening for LAML cancer and demonstrates the effectiveness of combining multiple factors in the drug discovery process. These findings have significant implications for the future of precision oncology and the development of more effective treatments for LAML cancer patients

Future Work

- In the future, it's expected that availability of several more data samples could help improve the accuracy of our model.
- This study consisted of looking at LAML, but the model can be used to test for any cancer and drug that comes from the databases.
- The Light Gradient Boosting Machine proved to be the best model, but in the future other regressors may have better results as data changes over time.

References

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